

T&M Toolkit: Options Available

The cost of a typical T&M tool kit for RF testing may vary from a few thousand dollars to over a million dollars depending upon the test requirement. For example, if it is a simple test requirement related to an undergraduate-course microwave laboratory of an educational institution (which may not require very sophisticated high-precision, high-performance test equipment), the combination of a general-purpose spectrum analyser and a signal generator will serve the purpose. But if the requirement is to test a more sophisticated system for satellite or radar subsystems, the test requirements are more stringent and the laboratory may comprise a VNA, performance signal analyser, microwave vector signal generator and many more equipment to supplement the testing needs.

There are a growing number of toolkit options available for RF engineers today. Emerging technologies require a flexible platform to keep pace with emerging communication standards. Unlike traditional RF signal generators and analysers that contain fixed features for standards such as GPS, wireless LAN (WLAN) and fixed WiMAX, virtual instrumentation provides equivalent measurement functionality while maintaining 'Open Source' software architecture.

Wireless standards toolkits provide a fully-flexible solution to both design and test of wireless devices. Because all waveforms are created and analysed in an Open Source software environment, one can quickly and easily introduce signal impairments, simulate wireless channel conditions and even apply custom filters. Combined with the general-purpose modulation toolkit, GPS, WLAN and fixed WiMAX toolkits can be used with both mixed-signal and RF instrumentation. As a result, engineers designing GPS, WLAN and WiMAX devices have access to a new and fully-flexible design tool for these devices. Many vendors have come up with cost-effective toolkits in the market, which range from \$2000 to \$6000 per licence.

Impact of new technologies

Experts believe that new technologies in the T&M industry will act as enablers for new developments in the RF domain. RF designers will be able to develop and test new wireless standards that require more accurate and repeatable set of measurements. It will also help them to reduce the design cycle by reducing the development time and hence time-to-market.

Vishal explains, "For example, the new analyser platform has brought down noise floor close to thermal-noise floor of -174 dBm. New techniques in measurement science like NVNA and X-parameter will help RF designers develop more accurate and more efficient amplifiers and RF circuits and hence reduce the overall power requirements of the RF systems without compromising on performance. The new handheld and low-cost instruments allow RF engineers to perform accurate measurements and pin-point the problems in the field. This will, in turn, enable field engineers to troubleshoot the problems on the go without bringing the modules back to repair centres and thus reduce the turnaround time."

Tarun believes that these new technologies will help the designer in two ways: "First, it will help to rapidly prototype complex RF subsystems using off-the-shelf hardware. High-bandwidth I/O combined with scalable processing will allow users to build prototypes very quickly, reducing the design cycle and time to market the product. Second, the testing can be comprehensive, resulting in better-quality products. The measurements that require processing of a huge amount of data can be done much faster.

"For example, PXIe backplane's peer-to-peer streaming feature enables users to combine RF modules like VSA and VSG with FPGA-based processor modules, thus integrating acquisition, analysis and generation for high-bandwidth and real-time operation."

What's in the offing?

A lot of interesting developments are underway. A few of these are listed below:

1. The industry is working towards high-performance test instruments. It is also working towards general-

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